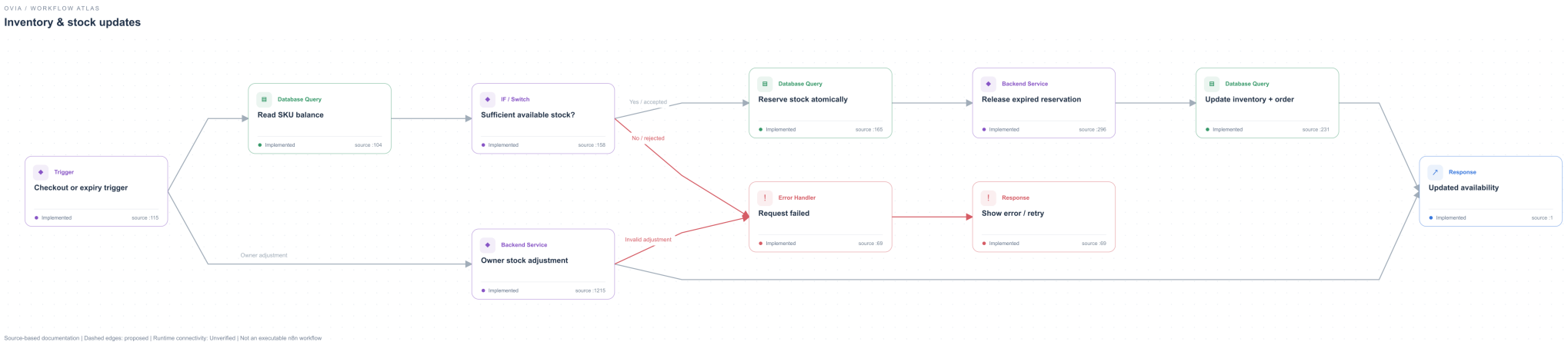
Inventory & stock updates

Architecture overview. Readable node and connection sheets follow.



Inventory & stock updates

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Checkout or expiry trigger

Reservation runs within checkout order creation; expiry runs via protected cron.

src/server/commerce.ts:115

Next -> #2 (p2); Owner adjustment -> #10 (p4)

#02 / Database Query / Implemented

Read SKU balance

inventory.id equals variant SKU; available quantity is loaded or derived from product variants.

src/server/commerce.ts:104

Next -> #3 (p2)

#03 / IF / Switch / Implemented

Sufficient available stock?

Reject overselling with HTTP 409 before updating the order.

src/server/commerce.ts:158

Yes / accepted -> #4 (p2); No / rejected -> #8 (p3)

#04 / Database Query / Implemented

Reserve stock atomically

Balance decrement and order insert share the MongoDB transaction.

src/server/commerce.ts:165

Next -> #5 (p3)

Inventory & stock updates

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / Backend Service / Implemented

Release expired reservation

Expiry code releases nonfinal inventory once; cancellation has its own release path.

src/server/commerce.ts:296

Next -> #6 (p3)

#06 / Database Query / Implemented

Update inventory + order

inventoryReleased protects repeated release.

src/server/commerce.ts:231

Next -> #7 (p3)

#07 / Response / Implemented

Updated availability

Catalog overlays inventory on variant stock.

src/components/product-detail.tsx:1

END: no outgoing connection

#08 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #9 (p4)

Inventory & stock updates

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#10 / Backend Service / Implemented

Owner stock adjustment

PATCH /api/admin/inventory updates inventory, product variant stock, inventoryMovements and audit.

src/server/admin-service.ts:1215

Next -> #7 (p3); Invalid adjustment -> #8 (p3)

Inventory & stock updates

Complete directed connection register / no omitted success, failure or proposed edges

#01 Checkout or expiry trigger -> #02 Read SKU balance | Next

#02 Read SKU balance -> #03 Sufficient available stock? | Next

#03 Sufficient available stock? -> #04 Reserve stock atomically | Yes / accepted

#04 Reserve stock atomically -> #05 Release expired reservation | Next

#05 Release expired reservation -> #06 Update inventory + order | Next

#06 Update inventory + order -> #07 Updated availability | Next

#03 Sufficient available stock? -> #08 Request failed | No / rejected | FAILURE

#08 Request failed -> #09 Show error / retry | Next | FAILURE

#01 Checkout or expiry trigger -> #10 Owner stock adjustment | Owner adjustment

#10 Owner stock adjustment -> #07 Updated availability | Next

#10 Owner stock adjustment -> #08 Request failed | Invalid adjustment | FAILURE

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.